

L²-Norm Two-Sample Mean Tests Under Compound Symmetry: Power Saturation, Decorrelation, and Signal Alignment

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The problem, and a puzzle

In high-dimension, low-sample-size (HDLSS) data with $p \gg n$ we test $H_0: \mu_1 = \mu_2$ against $H_1: \mu_1 \neq \mu_2$, where Hotelling's T^2 is undefined because the pooled covariance is singular. Three L^2 -norm tests avoid inversion by building on $\|\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2\|^2$: **Bai-Saranadasa (BS)** [1], **Chen-Qin (CQ)** [2] and **Srivastava-Du (SD)** [3].

An initial grid study (three covariances, the null plus three alternatives, $B = 1000$) found BS and CQ interchangeable and dominant under identity and AR(1). However, *all three looked powerless under compound symmetry (CS) at $\rho = 0.5$.*

BS power, sparse signal, $n = 20$	$p = 200$	500	2000	10 000
Identity	.308	.609	.984	1.000
Compound symmetry ($\rho = 0.5$)	.084	.081	.089	.106
its own null rate	.080	.066	.080	.085

Is that a failure of the tests, or a property of the covariance?

The governing quantity is $\text{tr}(\Sigma^2)$

For BS/CQ with $n_1 = n_2 = n$, the first-order local power is

$$\text{power} \approx \Phi(-z_\alpha + \text{SNR}), \quad \text{SNR} = \frac{n \|\delta\|^2}{2\sqrt{2} \sqrt{\text{tr} \Sigma^2}}, \quad \delta = \mu_1 - \mu_2 \quad (1)$$

Covariance enters only through $\text{tr}(\Sigma^2)$. With equal-norm signals, $\|\delta\|^2 = 0.0125 p$, and $\lambda_1 = 1 + (p-1)\rho$

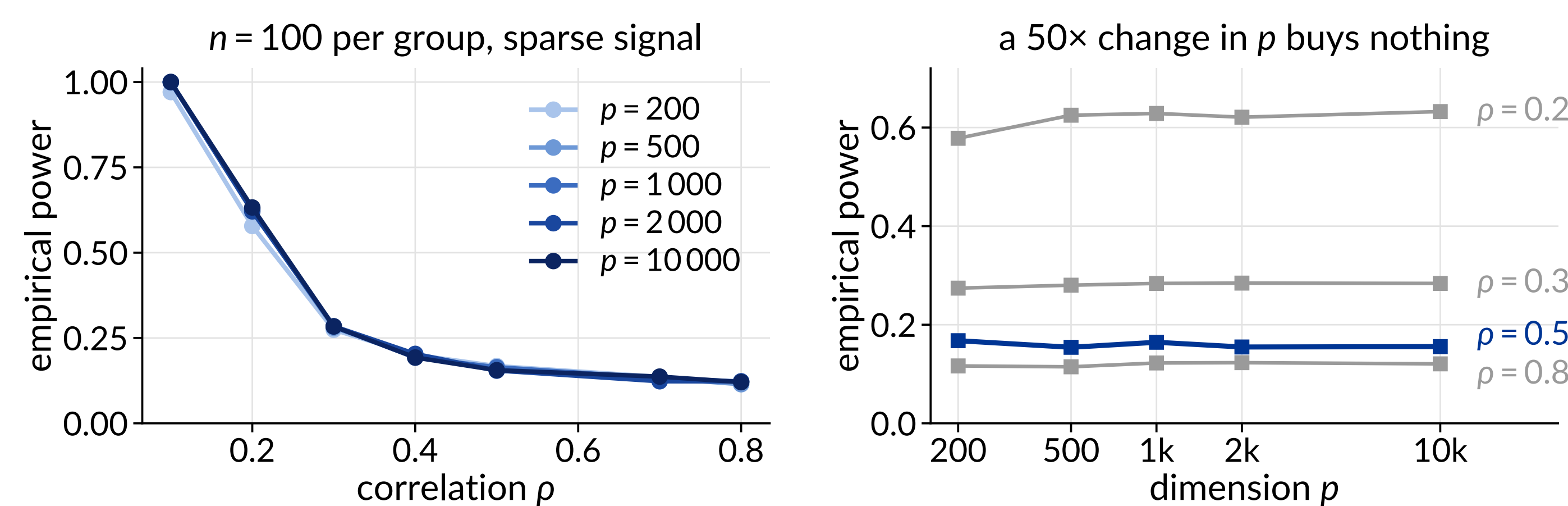
Σ	$\text{tr} \Sigma^2$	SNR $\propto$
Identity	p	$n\sqrt{p}$
AR(1), $\rho = 0.5$	$\frac{1+\rho^2}{1-\rho^2} p = \frac{5}{3}p$	$0.77 n\sqrt{p}$
CS, ρ	$\lambda_1^2 + (p-1)(1-\rho)^2 \approx \rho^2 p^2$	n

Under compound symmetry, the standard deviation of the statistic grows like p rather than $\sqrt{p}$, cancelling the linear growth of the signal:

$$\text{SNR}_{\text{CS}} = \frac{n \cdot 0.0125 p}{2\sqrt{2} p p} = \frac{0.0125 n}{2\sqrt{2} \rho} \quad \text{free of } p \text{ for fixed } \rho > 0 \quad (2)$$

Not powerless, but saturated. Only the dimension-driven gain is lost. At first order, 50% power at $\rho = 0.5$ needs $n \approx 186$ per group, far beyond the grid.

The dimension-driven gain does saturate



Raw BS power under CS, $B = 5000$ per cell. Left: the five dimensions are indistinguishable. Right: power against p on a log scale at fixed ρ .

At $n = 100$ the curves for $p = 500$ and $p = 10000$ are nearly identical (0.625 vs. 0.632 at $\rho = 0.2$; 0.154 vs. 0.156 at $\rho = 0.5$). Under identity, power instead climbs from .609 to 1.000 over that range of p , even at $n = 20$ (first table).

The null is not Gaussian either

Gaussian limits need $\text{tr}(\Sigma^4) = o(\text{tr}^2(\Sigma^2))$ [2]. Under fixed- ρ CS this fails maximally

$$\frac{\text{tr}(\Sigma^4)}{(\text{tr}(\Sigma^2))^2} \rightarrow \frac{(pp)^4}{((pp)^2)^2} = 1, \quad \text{equivalently} \quad \frac{\lambda_1^2}{\text{tr}(\Sigma^2)} \rightarrow 1 \quad (3)$$

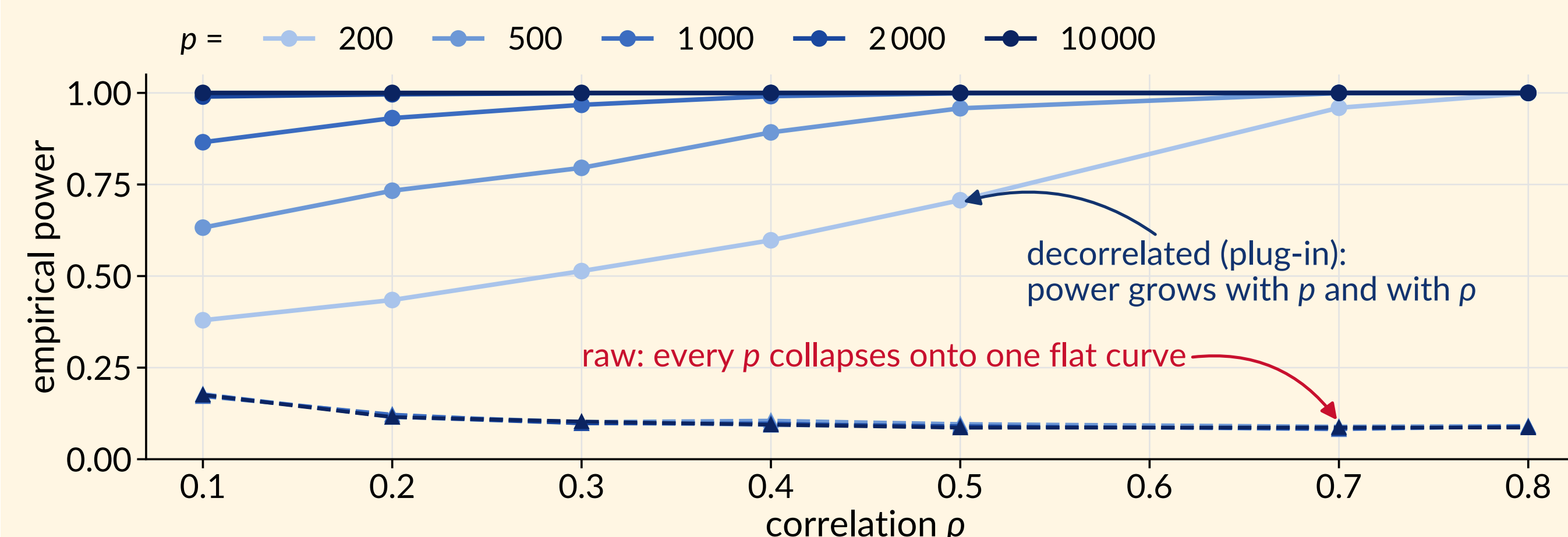
Hence, fixed- ρ CS is an **extreme strongly spiked eigenvalue model** [4]. Its standardized null tends to $(\chi_1^2 - 1)/\sqrt{2}$, and the normal 5% cutoff then gives size $\Pr(\chi_1^2 > 1 + 1.645\sqrt{2}) = 0.0682$.

Decorrelation: a closed-form $O(np)$ fix

CS has a known rank-one eigensystem, so the whitening map is explicit:

$$\Sigma^{-1/2} = a\mathbf{P}_\perp + b\mathbf{P}_1, \quad a = (1 - \rho)^{-1/2}, \quad b = \lambda_1^{-1/2},$$

applied per row as $x \mapsto a(x - \bar{x}\mathbf{1}) + b\bar{x}\mathbf{1}$ in $O(np)$: *no $p \times p$ object is ever formed*. The plug-in $\hat{\rho}$ (mean off-diagonal sample correlation) also costs $O(np)$.

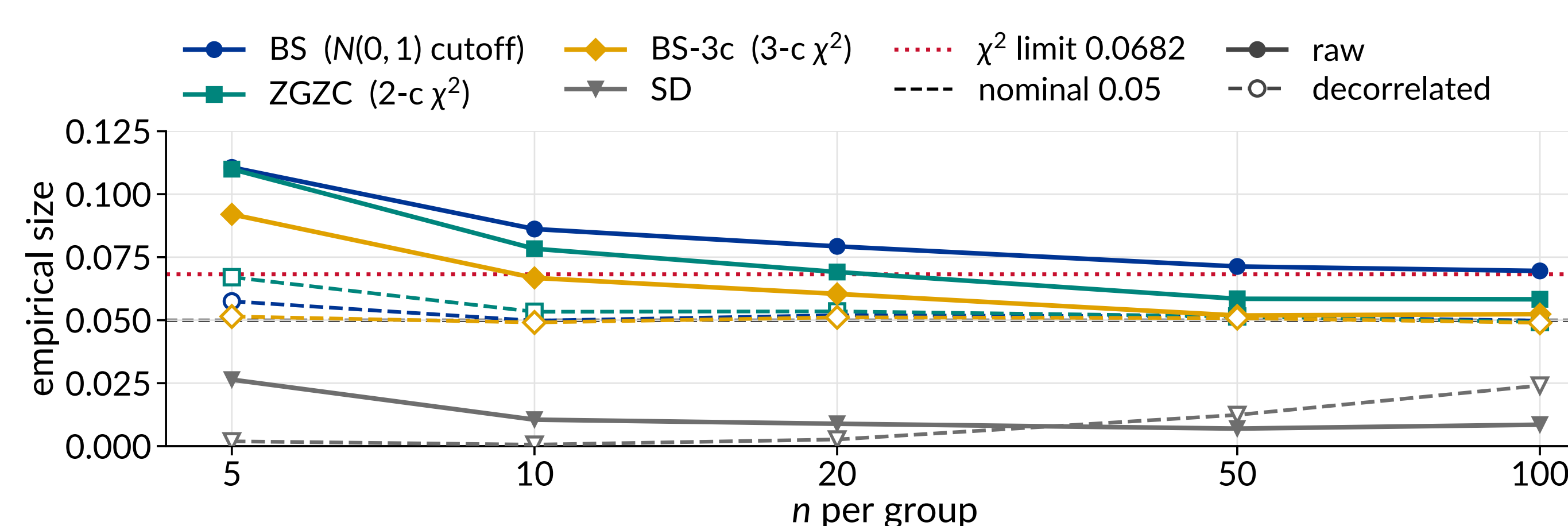


BS power, $n = 20$ per group, sparse signal, $B = 5000$: raw (dashed) vs. plug-in decorrelated (solid). Whitening restores an $O(n\sqrt{p})$ SNR through the orthogonal component of the signal.

At $p = 10000$ (sparse signal, averaged over ρ), whitening lifts power from 0.122 to 0.898 at $n = 5$ (0.71–1.00 across ρ), and to 1.000 at every ρ by $n = 20$. Over 210 alternative cells, predicted power matches simulation (MAE 0.0032, $r = 0.9998$).

The plug-in is as good as the oracle (0.898 vs. 0.902 in the same setting). For large p , $\text{sd}(\hat{\rho}) \approx \rho(1 - \rho)/\sqrt{n - 1}$ (mean observed/predicted ratio 1.019), and only $b/a = (1 - \hat{\rho})^{1/2} \lambda_1^{-1/2}$ enters the standardized statistic, so error in $\hat{\rho}$ costs little.

Two distinct problems: calibration and SNR



Size at $p = 10000$, averaged over ρ ; $B = 5000$ per cell ($\text{SE} \approx 0.003$). ZGZC [5] and BS-3c [6] are normal-reference tests (2- and 3-cumulant χ^2 approximations to the Gaussian-data null); all tests via HDNRA [7].

Raw BS/CQ size tends to 0.0682, not 0.05. Normal-reference tests bring the size near nominal by $n = 50$ but not the power: at $n = 100$, power minus size is BS 0.291, ZGZC 0.264, BS-3c 0.242. Recalibration moves the cutoff, not $\|\delta\|/\sqrt{\text{tr} \Sigma^2}$. Whitening fixes both. SD stays conservative, a separate p/n -driven problem.

Signal alignment, the hidden variable

Decompose the mean difference along the common-factor direction $u = \mathbf{1}_p/\sqrt{p}$

$$\delta = c_1 u + \delta_\perp, \quad \gamma = \frac{c_1^2}{\|\delta\|^2} \in [0, 1]$$

For an equal shift on a fraction s of coordinates, $\gamma = s$ exactly, so the sparse, moderate and dense designs ($s = 5\%$, 20%, 100%) confounded *sparsity* with *alignment*. Oracle whitening gives

$$\|\tilde{\delta}\|^2 = \|\delta\|^2 \left[\frac{\gamma}{\lambda_1} + \frac{1 - \gamma}{1 - \rho} \right],$$

so whitening helps through $\delta_\perp$ and hurts only as $\gamma \rightarrow 1$. Keeping the spike explicit and treating the bulk as normal gives a finite- n rule for the trace-centred statistic $M = \|\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2\|^2 - 2 \text{tr} \Sigma/n$

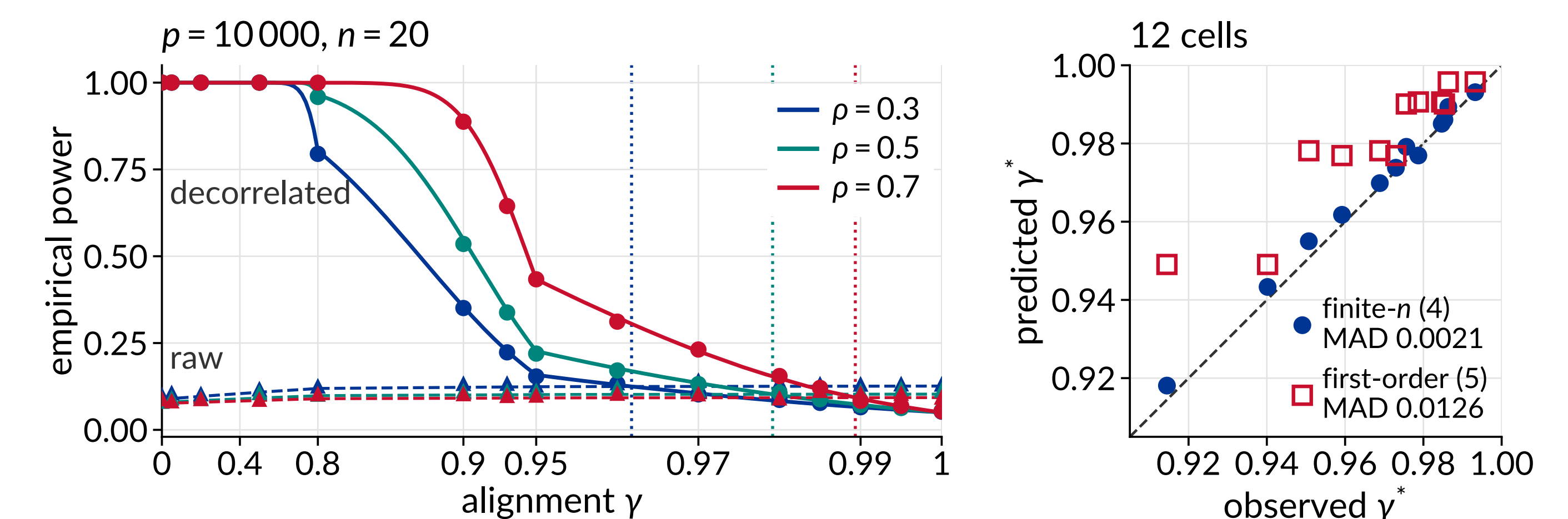
$$M \stackrel{d}{\approx} a_u [(Z + m)^2 - 1] + (1 - \gamma)E + s_b V, \quad (4)$$

with Z, V independent $N(0, 1)$, $E = \|\delta\|^2$, $a_u = 2\lambda_1/n$, $a_b = 2(1 - \rho)/n$, $m^2 = \gamma E/a_u$ and $s_b^2 = 2(p - 1)a_b^2 + 4a_b(1 - \gamma)E$.

The crossover

Equating first-order raw and whitened SNRs gives, in closed form,

$$\gamma_{\text{SNR}}^*(p, \rho) = \frac{(1 - \rho)^{-1} - [1 + (p - 1)\rho^2]^{-1/2}}{(1 - \rho)^{-1} - [1 + (p - 1)\rho]^{-1}} = 1 - \frac{1 - \rho}{\rho\sqrt{p}} + o(p^{-1/2}) \quad (5)$$



Left: power against γ (axis stretched above 0.8); markers simulated (plug-in), lines the finite- n rule (4), dotted rules its crossings. Right: predicted vs. observed γ^* over $p \in \{2000, 10000\}$, $n \in \{20, 50\}$, $\rho \in \{0.3, 0.5, 0.7\}$. The first-order (5) sits above every crossing (gap shrinking in n); (4) tracks them.

Take-aways

- 1. Saturation, not failure.** Fixed positive CS cancels the dimension-driven power gain exactly. SNR then depends on n and ρ only.
- 2. Calibration is not power.** Normal-reference tests fix the size (for $n \geq 50$) but not the power; whitening recovers the SNR, and here the size too.
- 3. The trade-off is one-sided** (alignment grid, $n \in \{20, 50\}$). Maximum gain from decorrelation 0.858–0.921, maximum loss 0.040–0.171: harm is confined to $\gamma \approx 1$, where raw power is only 0.09–0.23.

Key references

- [1] Bai & Saranadasa (1996) *Stat. Sinica* **6**, 311–329.
- [2] Chen & Qin (2010) *Ann. Statist.* **38**, 808–835.
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- [4] Aoshima & Yata (2018) *Stat. Sinica* **28**, 43–62.
- [5] Zhang, Guo, Zhou & Cheng (2020) *JASA* **115**, 1011–1027.
- [6] Zhang & Zhu (2022) *J. Nonparametr. Stat.* **34**, 58–76.
- [7] Nie, Zhu, Zhang & Zhang (2024) HDNRA, R package, CRAN.